UofI Hand-Built Robot

The UofI Hand-Built Robot is a ROS 2-based autonomous robot system built by students at the University of Illinois. The robot uses a mecanum-wheel drive base for omnidirectional movement, a fisheye camera for computer vision, and a 5-DOF xArm manipulator for picking up objects.

The system is controlled by a finite state machine (FSM) that orchestrates the entire object collection pipeline: searching for objects, selecting the best target, approaching it, grasping it with the arm, finding the drop box, and delivering the object. The vision system uses YOLO with TensorRT inference to detect objects, while the navigation system uses a novel "whisker" proximity sensing approach that casts virtual rays to estimate distances to objects.

The project is designed for student learning and experimentation. It includes multiple debugging GUIs, tuning tools, and supports the Foxglove visualization platform for remote monitoring. The entire system runs on an NVIDIA Jetson computer mounted on the robot, making it a fully autonomous platform.

Getting Started

1. Set up your ROS 2 workspace by sourcing the setup file: `bash source /opt/ros/humble/setup.bash # Linux # or on Windows with ROS 2 installed: call C:\dev\ros2_humble\local_setup.bat`
2. Build the workspace: `bash cd C:\GITHUB Repos\UofIHandBuiltRobot colcon build --symlink-install`
3. Source the workspace: `bash source install/setup.bash # Linux call install/setup.bat # Windows`
4. Launch the full system with GUI: `bash ros2 launch system_manager_package robot_bringup.launch.py launch_gui:=true`

You should see the Robot State GUI appear showing the camera feed, WSKR overlay, and current robot state. The robot is now ready to accept commands.

Installation

System Requirements

- **Operating System**: Ubuntu 22.04 (recommended) or Windows 10/11
- **ROS 2 Distribution**: Humble Hawksbill

- **Python**: 3.10 or later
- **Hardware**: NVIDIA Jetson (Orin, Xavier, or Nano) with GPU support
- **Camera**: USB camera supporting MJPEG at 1920x1080@30fps
- **Arduino**: Arduino-compatible board for motor control

ROS 2 Installation

Ubuntu 22.04

```
# Install ROS 2 Humble
sudo apt update
sudo apt install ros-humble-desktop-full

# Install additional packages
sudo apt install python3-pip python3-venv
pip3 install opencv-python-headless numpy pillow

# Source ROS 2
source /opt/ros/humble/setup.bash
```

Windows

1. Download ROS 2 Humble from https://github.com/ros2/ros2/releases
2. Run the installer and follow the prompts
3. Add ROS 2 to your PATH: `cmd setx PATH "%PATH%;C:\dev\ros2_humble\bin"`

Building the Workspace

```
# Navigate to the project root
cd C:\GITHUB Repos\UofIHandBuiltRobot

# Build with colcon
colcon build --symlink-install

# Source the workspace
source install/setup.bash  # Linux
call install\setup.bat     # Windows
```

Foxglove Installation on Jetson

Foxglove is a modern robotics visualization tool that provides real-time data streaming and debugging capabilities.

Step 1: Install Foxglove Studio

```
# Download Foxglove Studio for ARM64 (Jetson)
wget https://github.com/foxglove/studio/releases/latest/download/FoxgloveStudio-linux-arm64.deb

# Install the package
sudo apt install ./FoxgloveStudio-linux-arm64.deb
```

Step 2: Install Foxglove Bridge

The foxglove_bridge package is already included as a dependency in the `system_manager_package`. To verify it's installed:

```
ros2 pkg list | grep foxglove
```

Step 3: Launch Foxglove Bridge

```
# Start the bridge (runs alongside your robot nodes)
ros2 launch foxglove_bridge foxglove_bridge_launch.xml port:=8765
```

Step 4: Connect Foxglove Studio

1. Open Foxglove Studio on your Jetson or remote computer
2. Click "Open connection" → "Foxglove WebSocket"
3. Enter the WebSocket URL: `ws://<jetson-ip-address>:8765`
4. Click "Open"

Step 5: Load the WSKR Dashboard Layout

1. In Foxglove Studio, go to File → Open
2. Select `foxglove_files/wskr_dashboard_remote.json` from the project
3. The dashboard will load with pre-configured panels for: - Whisker fan visualization - Camera feed with ArUco detection - cmd_vel plots - Heading and tracking mode indicators - Model selection and speed control

Troubleshooting Foxglove

- **Connection refused**: Ensure the bridge is running and the port is not blocked by firewall
- **No data showing**: Verify ROS 2 topics are being published with `ros2 topic list`
- **Slow performance**: Reduce image throttle rates in the bridge configuration

Dependencies

Install Python dependencies:

```
pip install -r requirements.txt
```

Key dependencies include: - `opencv-python`: Computer vision operations - `numpy`: Numerical computations - `pillow`: Image processing - `tensorrt`: GPU-accelerated inference (Jetson only) - `ultralytics`: YOLO model inference

Project Structure

The project is organized as a ROS 2 workspace with multiple packages in the `src/` directory:

```
UofIHandBuiltRobot/
├── src/
│   ├── arduino/                      # Arduino serial bridge for motor control
│   │   ├── arduino/
│   │   │   └── mecanum_serial_bridge.py
│   │   └── launch/
│   │       └── arduino.launch.py
│   │
│   ├── gstreamer_camera/             # Camera driver using GStreamer
│   │   ├── gstreamer_camera/
│   │   │   └── gst_cam_node.py
│   │   └── launch/
│   │       └── gstreamer_camera.launch.py
│   │
│   ├── system_manager_package/       # Central FSM and GUI tools
│   │   ├── gui/
│   │   │   └── robot_gui.py          # Main debugging GUI
│   │   ├── src/
│   │   │   └── state_manager.py      # Finite state machine
│   │   ├── system_manager_package/
│   │   │   └── constants.py          # All tunable parameters
│   │   └── launch/
│   │       ├── robot_bringup.launch.py
│   │       └── sys_manager.launch.py
│   │
│   ├── wskr/                         # Whisker-based navigation
│   │   ├── wskr/
│   │   │   ├── wskr_autopilot.py     # MLP-based reactive controller
│   │   │   ├── approach_action_server.py
│   │   │   ├── wskr_floor_node.py    # Floor detection
│   │   │   ├── wskr_range_node.py    # Whisker ray casting
│   │   │   └── dead_reckoning_node.py
│   │   └── launch/
│   │       └── wskr.launch.py
│   │
│   ├── wskr_search_behavior/         # Search and wander behavior
│   │   └── wskr_search_behavior/
│   │       └── search_supervisor.py
│   │
│   ├── vision_processing_package/    # YOLO inference and object selection
│   │   ├── src/
│   │   │   ├── process_object_vision.py
│   │   │   ├── object_selection.py
│   │   │   └── bbox_to_xyz_service_2D.py
│   │   └── launch/
│   │       └── vision_processing.launch.py
│   │
│   ├── xarm_object_collector_package/ # xArm manipulator control
```

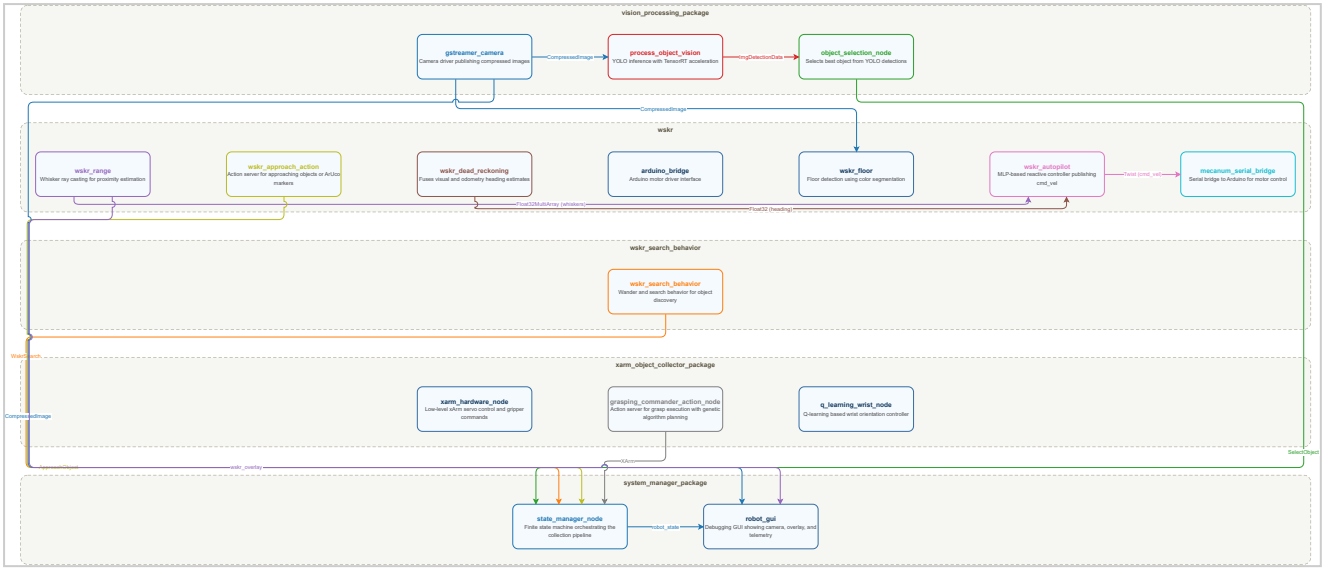
```
│   │   ├── src/
│   │   │   ├── xarm_hardware_node.py
│   │   │   ├── Object_collector_action_server.py
│   │   │   ├── genetic_algorithm.py
│   │   │   └── controller_class.py
│   │   └── launch/
│   │       └── xarm_object_collector_ga.launch.py
│   │
│   ├── utilities/                    # Debugging and tuning tools
│   │   ├── utilities/
│   │   │   ├── wskr_dashboard.py     # Combined control dashboard
│   │   │   ├── floor_tuner.py        # Floor detection tuner
│   │   │   ├── heading_tuner.py      # Heading estimation tuner
│   │   │   ├── mecanum_teleop.py     # Manual teleoperation
│   │   │   └── robot_control_panel.py
│   │   └── launch/
│   │       └── wskr_foxglove.launch.py
│   │
│   └── robot_interfaces/             # Custom ROS 2 message/action definitions
│
├── foxglove_files/                   # Foxglove dashboard layouts
│   ├── wskr_dashboard_remote.json
│   └── wskr_default.json
│
├── requirements.txt                  # Python dependencies
└── docs/                             # Documentation output
```

Package Summary

Package	Purpose
`arduino`	Serial communication with Arduino motor controller
`gstreamer_camera`	Camera driver publishing compressed images
`system_manager_package`	FSM orchestrator and debugging GUI
`wskr`	Whisker-based proximity sensing and autopilot
`wskr_search_behavior`	Search and wander behavior for object discovery
`vision_processing_package`	YOLO object detection and selection
`xarm_object_collector_package`	xArm manipulator control with GA planning
`utilities`	Development and debugging tools
`robot_interfaces`	Custom ROS 2 message and action definitions

robot_bringup.launch.py — Full System Architecture

This diagram shows the complete node graph when running `robot_bringup.launch.py`. The launch file includes four sub-launch files: sys_manager.launch.py (state manager), xarm_object_collector_ga.launch.py (xArm control), wskr.launch.py (navigation), and search_behavior.launch.py (search behavior). Each node is grouped by its package. Topics flow between nodes for sensor data, commands, and status updates.



All ROS 2 nodes launched by robot_bringup.launch.py, grouped by package. Topics connect nodes for data flow.

Core Concepts

The Finite State Machine (FSM)

The robot's behavior is controlled by a **Finite State Machine** implemented in `state_manager.py`. The FSM manages the entire object collection pipeline through a series of states:

```
IDLE ──(command)──► SEARCH ──► SELECT ──► APPROACH_OBJ ──► GRASP
                     ▲                                       │
                     │                                       ▼
                     │                                    FIND_BOX
                     │                                       │
                     │                                       ▼
                     ◄──────────── DROP ◄──────────── APPROACH_BOX
```

State Descriptions

State	Purpose	Next State
IDLE	Robot is stationary, waiting for commands	SEARCH (on command)
SEARCH	Wander while looking for objects using YOLO	SELECT (object found)
SELECT	Pick the best object from current detections	APPROACH_OBJ
APPROACH_OBJ	Drive toward the selected object	GRASP (proximity reached)
GRASP	Pick up object with xArm	FIND_BOX (success)
FIND_BOX	Wander while looking for ArUco marker on drop box	APPROACH_BOX
APPROACH_BOX	Drive toward the drop box	DROP (proximity reached)
DROP	Open gripper to release object	SEARCH (loop continues)
STOPPED	Emergency stop - ignores most commands	IDLE (on "idle" command)
ERROR	Error state - requires manual intervention	IDLE (on "idle" command)

How the FSM Works

1. **Commands**: Send state names to `/robot_command` topic (e.g., "search", "stop")
2. **Transitions**: Each state has a delay before its handler executes
3. **Actions**: States dispatch ROS 2 action goals to subsystems (search, approach, grasp)
4. **Feedback**: Action feedback updates progress; results trigger next state
5. **Cancellation**: STOPPED/ERROR states cancel all in-flight actions

Publishing Commands

```
# Start searching for objects
ros2 topic pub /robot_command std_msgs/msg/String "{data: 'search'}"

# Stop the robot
ros2 topic pub /robot_command std_msgs/msg/String "{data: 'stop'}"

# Return to idle
ros2 topic pub /robot_command std_msgs/msg/String "{data: 'idle'}"
```

Whisker-Based Navigation

The WSKR (Whisker-based Sensorimotor Control for Robots) system uses a **fisheye camera** to estimate distances to objects:

1. **Floor Detection**: Identifies floor pixels using color segmentation
2. **Ray Casting**: Casts 11 virtual "whisker" rays across the camera's field of view
3. **Distance Estimation**: Measures distance to first non-floor pixel along each ray
4. **Proximity Sensing**: Provides 11 distance values (in mm) representing obstacles

The whisker system works like an insect's antennae - providing proximity information without requiring depth sensors.

Dead Reckoning Fusion

The robot fuses two heading estimates:

- **Visual Heading**: Computed from bounding box position in camera frame
- **Odometry Heading**: Computed from wheel encoder data

When the visual heading becomes unreliable (object leaves camera view), the system switches to dead reckoning. When the object reappears, it switches back to visual tracking.

Autopilot (MLP Controller)

The autopilot uses a **Multi-Layer Perceptron (MLP)** trained with reinforcement learning:

- **Input**: 11 whisker distances + 11 target whisker distances + 1 heading angle = 23 values
- **Output**: cmd_vel (linear x, linear y, angular z)
- **Training**: PPO (Proximal Policy Optimization) in simulation
- **Inference**: Runs at 10 Hz on the Jetson GPU

Genetic Algorithm Trajectory Planning

The xArm uses a **Genetic Algorithm** to plan collision-free trajectories:

1. **Chromosome**: Sequence of joint actions (step size = 2°)
2. **Fitness**: Minimizes distance to goal + penalizes non-orthogonal approach
3. **Evolution**: 150 individuals × 100 generations
4. **Output**: Smooth joint trajectory for the 5-DOF arm

Key Terminology

Term	Definition
Whisker	A virtual ray cast from the camera to estimate distance
ArUco Marker	A fiducial marker (like a QR code) used for precise localization
YOLO	"You Only Look Once" - a real-time object detection neural network
TensorRT	NVIDIA's GPU inference optimizer for fast neural network execution
Dead Reckoning	Estimating position from wheel encoder data
Proximity	Distance to nearest obstacle along a whisker ray
Tracking Mode	Either "visual" (using camera) or "dead_reckoning" (using odometry)

Full System Bringup and Autonomous Collection

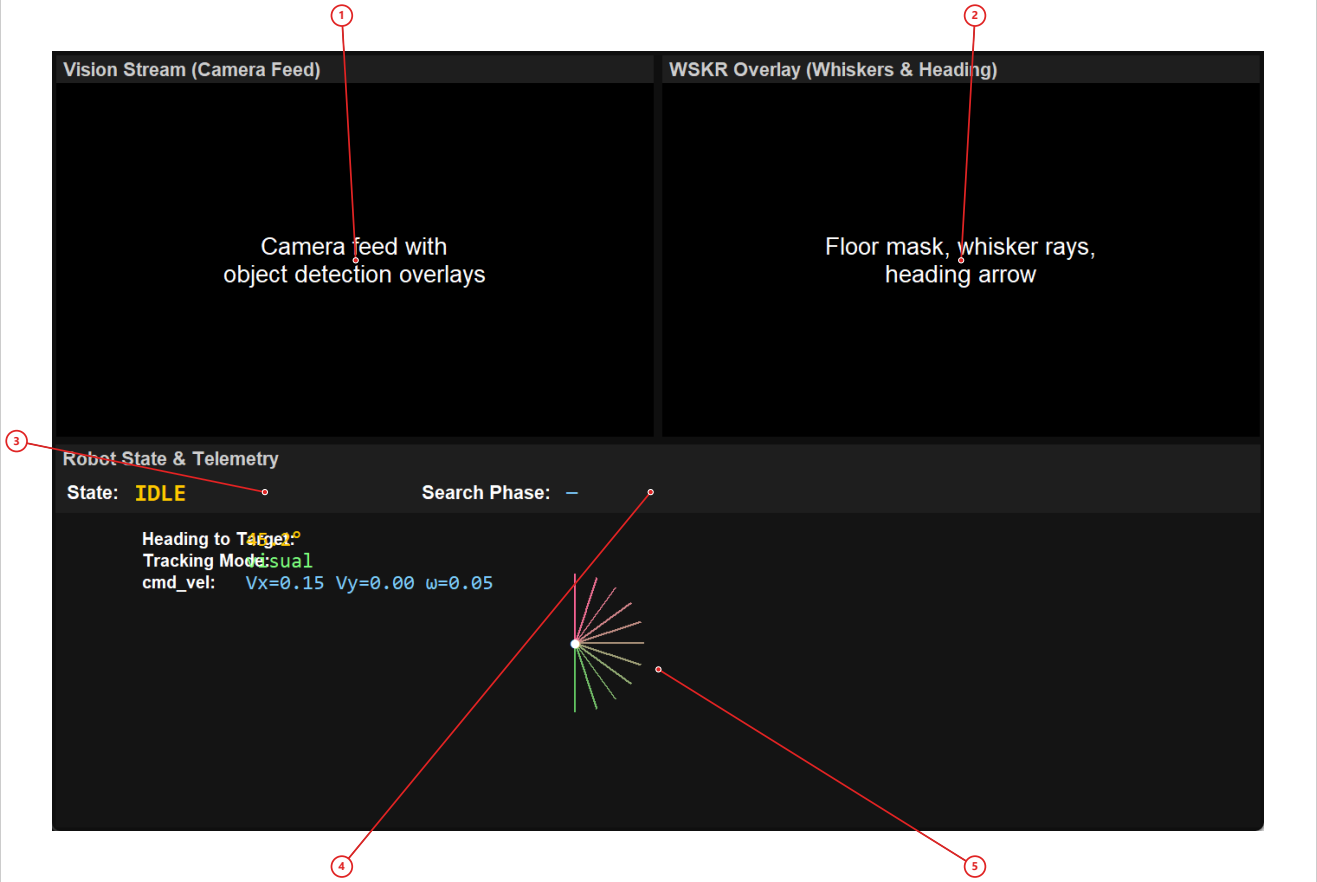
Launch the complete robot system and begin autonomous object collection. This workflow starts all nodes including the state machine, vision system, navigation, xArm control, and search behavior.

How to use

1. 1. Open a terminal and source your ROS 2 workspace: `bash source /opt/ros/humble/setup.bash cd C:\GITHUB Repos\UofIHandBuiltRobot source install/setup.bash`
2. 2. Launch the full system with the debugging GUI: `bash ros2 launch system_manager_package robot_bringup.launch.py launch_gui:=true` Wait for all nodes to start (about 10-15 seconds). You should see console output from each node.
3. 3. The **Robot State GUI** will open showing: - **Vision Stream** (top-left): Live camera feed with object detection overlays - **WSKR Overlay** (top-right): Floor mask, whisker rays, and heading arrow - **State & Telemetry** (bottom): Current robot state, search phase, heading, and cmd_vel
4. 4. Verify the robot is in **IDLE** state (shown in yellow in the state display).
5. 5. Start the autonomous collection by publishing to the command topic: `bash ros2 topic pub /robot_command std_msgs/msg/String "{data: 'search'}"`
6. 6. Watch the state transitions in the GUI: - **SEARCH**: Robot wanders while looking for objects - **SELECT**: Object detected, selecting best target - **APPROACH_OBJ**: Driving toward the object - **GRASP**: xArm picking up the object - **FIND_BOX**: Searching for the drop box ArUco marker - **APPROACH_BOX**: Driving to the drop box - **DROP**: Releasing the object - **SEARCH**: Loop repeats for continuous collection
7. 7. Monitor the telemetry panel for: - **Heading to Target**: Current heading angle in degrees - **Tracking Mode**: 'visual' or 'dead_reckoning' - **cmd_vel**: Current velocity commands (Vx, Vy, ω) - **Whisker Fan**: Visual representation of proximity sensors
8. 8. To stop the robot at any time: `bash ros2 topic pub /robot_command std_msgs/msg/String "{data: 'stop'}"`
9. 9. To resume from stopped: `bash ros2 topic pub /robot_command std_msgs/msg/String "{data: 'idle'}"`

Robot State GUI

Main debugging dashboard showing live camera feed, WSKR navigation overlay, and real-time robot telemetry.



Labeled controls and displays:

1. **Vision Stream panel** — Live camera feed with YOLO detection bounding boxes overlaid
2. **WSKR Overlay panel** — WSKR navigation overlay showing floor mask, whiskers, and heading
3. **Robot state display** — Current FSM state (IDLE, SEARCH, SELECT, APPROACH_OBJ, etc.)
4. **Search phase indicator** — Current search behavior phase (WANDER, LOOK, etc.)
5. **Telemetry canvas** — Numeric telemetry: heading, tracking mode, cmd_vel, whisker fan diagram

The **Robot State GUI** provides comprehensive debugging information for the autonomous collection pipeline. The **Vision Stream** panel (1) shows the live camera feed with YOLO detection bounding boxes overlaid in different colors. The **WSKR Overlay** panel (2) displays the floor detection mask, whisker proximity rays, heading arrow, and current tracking mode. The **State display** (3) shows the current FSM state (IDLE, SEARCH, SELECT, etc.) in yellow text. The **Search Phase** indicator (4) shows the current phase of the search behavior (WANDER, LOOK, etc.). The **Telemetry canvas** (5) at the bottom displays numeric values for heading to target, tracking mode, cmd_vel commands, and a visual whisker fan diagram showing proximity distances. Use this GUI to monitor robot behavior during autonomous operation and debug issues with vision, navigation, or state transitions.

WSKR Dashboard for Manual Control and Debugging

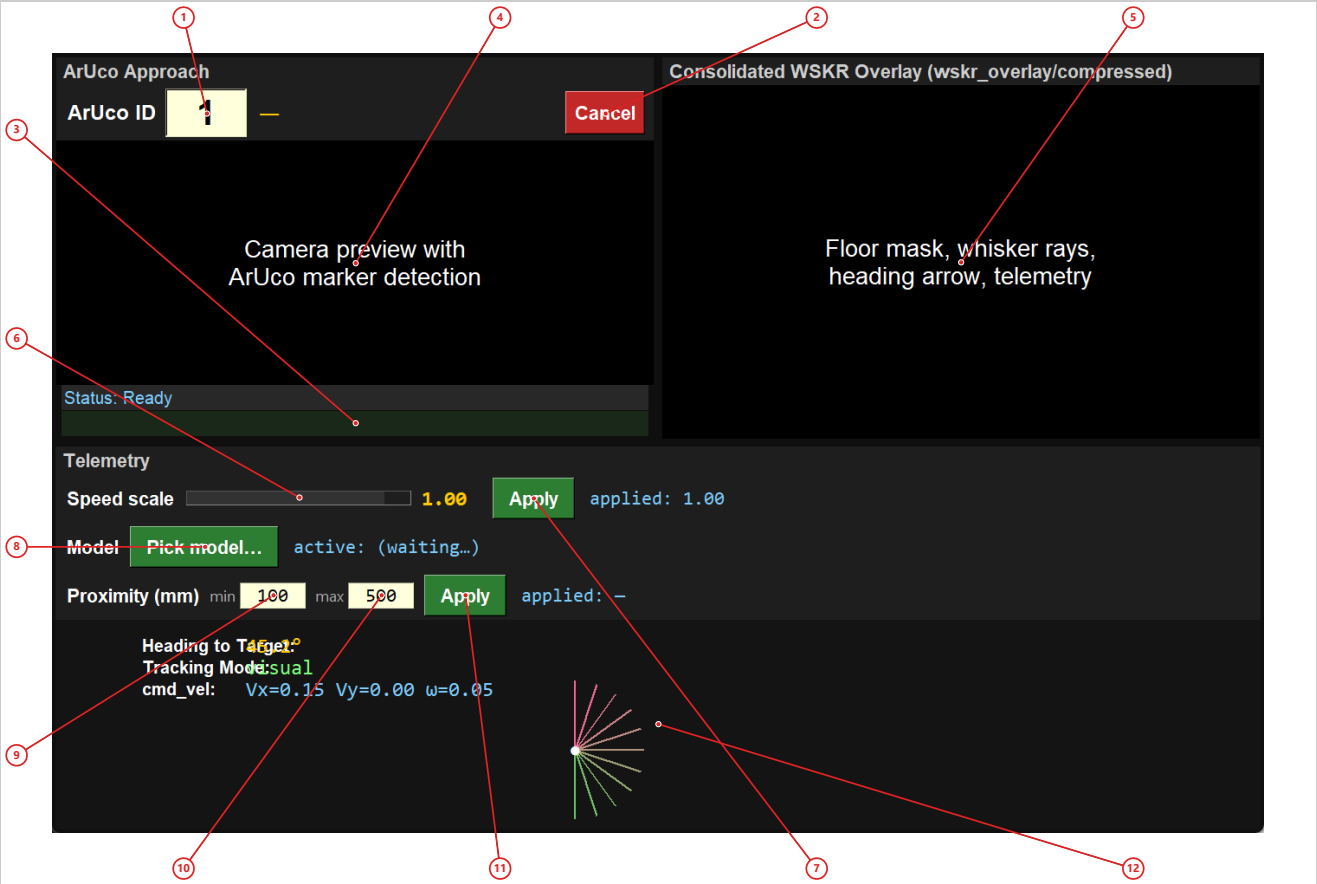
Use the WSKR Dashboard for manual control and debugging of the approach behavior. This GUI provides direct control over ArUco approach, speed scaling, model selection, and proximity parameters.

How to use

1. 1. Launch the WSKR Dashboard: `bash ros2 run utilities wskr_dashboard` The dashboard will open with three panels.
2. 2. **ArUco Approach Panel** (top-left): - Enter the ArUco marker ID in the entry field (1) - Click **Start Approach** (2) to begin approaching the marker - Monitor the **Status** label (3) for goal state updates - Watch the **Camera preview** (4) for live ArUco detection (yellow box = target marker) - Click **Cancel** to abort the approach
3. 3. **WSKR Overlay Panel** (top-right): - Shows the consolidated WSKR telemetry overlay - Displays floor mask (white = detected floor) - Shows whisker rays (colored lines = proximity distances) - Shows heading arrow and current tracking mode
4. 4. **Telemetry Panel** (bottom): - **Speed scale slider** (5): Adjust autopilot speed from 0.0 to 1.0 - Click **Apply** (6) to send the new speed setting - **Pick model** button (7): Select a different neural network model file - **Proximity min/max** entries (8, 9): Set distance thresholds for speed attenuation - Click **Apply** (10) to update proximity parameters - **Telemetry canvas** (11): Shows heading, tracking mode, cmd_vel, and whisker fan diagram
5. 5. To approach a toy object instead of an ArUco marker: - Ensure the vision system is detecting objects (check YOLO output) - Click **Approach Toy** button - The robot will approach the object selected by object_selection_node
6. 6. Monitor the feedback labels: - **Approach status**: Shows current goal state (ACTIVE, SUCCESS, ABORTED, etc.) - **Feedback row**: Shows live tracking mode, heading, and closest whisker distance during approach
7. 7. Tune parameters in real-time: - Adjust speed scale to make the robot move slower/faster - Change proximity limits to control when the robot slows down near objects - Switch models to test different trained policies

WSKR Dashboard

Combined control dashboard for ArUco approach, telemetry monitoring, and autopilot parameter tuning.



Labeled controls and displays:

1. **ArUco ID entry** — Enter the ArUco marker ID to approach (default: 1)
2. **Cancel button** — Cancel the current approach goal
3. **Approach status label** — Current approach goal state (Ready, Active, Success, Aborted)
4. **Camera preview** — Live camera feed with ArUco marker detection overlay
5. **WSKR Overlay panel** — Consolidated WSKR telemetry overlay with floor mask and whiskers
6. **Speed scale slider** — Adjust autopilot speed scale from 0.0 to 1.0
7. **Apply speed button** — Apply the selected speed scale to the autopilot
8. **Pick model button** — Select a different neural network model file
9. **Proximity min entry** — Minimum proximity distance for speed attenuation (mm)
10. **Proximity max entry** — Maximum proximity distance for speed attenuation (mm)
11. **Apply proximity button** — Apply the proximity limits to the autopilot
12. **Telemetry canvas** — Telemetry display: heading, tracking mode, cmd_vel, whisker fan

The **WSKR Dashboard** provides direct control over the approach behavior and real-time telemetry. The **ArUco ID entry** (1) lets you specify which ArUco marker to approach. Click **Start Approach** to begin, and **Cancel** (2) to abort. The **Status label** (3) shows the current goal state. The **Camera preview** (4) displays live video with ArUco marker detection (yellow box indicates the target marker). The **WSKR Overlay** (5) shows the consolidated navigation display. Use the **Speed scale slider** (6) to adjust autopilot speed from 0% to 100%, then click **Apply** (7). The **Pick model** button (8) opens a file dialog to select a different neural

network policy. The **Proximity min** (9) and **max** (10) entries set the distance thresholds for speed attenuation—click **Apply** (11) to update. The **Telemetry canvas** (12) shows heading to target, tracking mode, cmd_vel values, and a whisker fan diagram. This dashboard is ideal for testing approach behavior and tuning autopilot parameters.

Component Testing and Debugging

Test individual robot subsystems in isolation to verify correct operation and debug issues. This workflow covers testing the camera, vision, whisker system, state machine, and xArm independently.

How to use

1. **Testing the Camera**
2. Launch only the camera node: `bash ros2 launch gstreamer_camera gstreamer_camera.launch.py`
3. Verify the camera is publishing: `bash ros2 topic hz /camera1/image_raw/compressed` You should see ~10 Hz output.
4. View the camera feed: `bash ros2 run rqt_image_view rqt_image_view` Select `/camera1/image_raw/compressed` from the dropdown.

Testing Vision Processing

1. Launch camera and vision nodes: `bash ros2 launch vision_processing_package vision_processing.launch.py`
2. Check YOLO detections: `bash ros2 topic echo /vision/yolo/detections` Point the camera at objects to see detection messages.
3. Test object selection: `bash ros2 service call /select_object_service robot_interfaces/srv/SelectObject`

Testing the Whisker System

1. Launch WSKR nodes: `bash ros2 launch wskr wskr.launch.py`
2. Monitor whisker lengths: `bash ros2 topic echo /WSKR/whisker_lengths` Wave your hand in front of the camera to see distance changes.
3. Check floor detection: `bash ros2 topic hz /wskr_overlay/compressed`

Testing the State Machine

1. Launch only the state manager: `bash ros2 launch system_manager_package sys_manager.launch.py`
2. Monitor robot state: `bash ros2 topic echo /robot_state`

3. Send test commands: ```bash # Should transition to SEARCH ros2 topic pub /robot_command std_msgs/msg/String "{data: 'search'}"

Should transition to IDLE ros2 topic pub /robot_command std_msgs/msg/String "{data: 'idle'}"

Should transition to STOPPED ros2 topic pub /robot_command std_msgs/msg/String "{data: 'stop'}" ```

4. Verify state transitions appear in the echo output.

Testing the xArm

1. Launch xArm nodes: `bash ros2 launch xarm_object_collector_package xarm_object_collector_ga.launch.py`
2. Check xArm connection: `bash ros2 run xarm_object_collector_package test_xarm_connection.py`
3. Test gripper service: `bash # Open gripper ros2 service call /open_gripper_service std_srvs/srv/Trigger`

Testing the Autopilot

1. Launch WSKR with autopilot: `bash ros2 launch wskr wskr.launch.py`
2. Monitor cmd_vel output: `bash ros2 topic echo /WSKR/cmd_vel`
3. Provide synthetic whisker input (advanced): `bash ros2 topic pub /WSKR/whisker_lengths std_msgs/msg/Float32MultiArray "{data: [200.0, 200.0, 200.0, 200.0, 200.0, 200.0, 200.0, 200.0, 200.0, 200.0, 200.0]}"`

Testing Search Behavior

1. Launch search behavior: `bash ros2 launch wskr_search_behavior search_behavior.launch.py`
2. Send a search goal (requires action client): `python # Create a test script or use rclpy in Python`
3. Monitor search phase: `bash ros2 topic echo /WSKR/search_phase`

Configuration and Tuning

Parameter Configuration

All tunable parameters are centralized in `src/system_manager_package/system_manager_package/constants.py`. This single source of truth ensures consistent behavior across all nodes.

Key Parameter Groups

Camera Parameters

```
CAMERA_WIDTH = 1920                    # Capture resolution
CAMERA_HEIGHT = 1080
CAMERA_FPS = 30
CAMERA_PUBLISH_HZ = 10.0               # Throttled publish rate
```

WSKR Floor Detection

```
FLOOR_BLUR_KERNEL = 9                  # Gaussian blur size
FLOOR_COLOR_DIST_THRESH = 20           # LAB color distance threshold
FLOOR_GRADIENT_THRESH = 14             # Edge detection threshold
```

WSKR Whisker Range

```
WHISKER_MAX_RANGE_MM = 500.0           # Maximum ray-march distance
WHISKER_SAMPLE_STEP_MM = 1.0           # Sample granularity
```

Autopilot (MLP Controller)

```
AUTOPILOT_MAX_LINEAR_MPS = 0.40        # Max forward/strafe speed (m/s)
AUTOPILOT_MAX_ANGULAR_RPS = 0.698      # Max rotation (40 deg/s)
AUTOPILOT_SPEED_SCALE = 1.0            # Global output scaling [0,1]
AUTOPILOT_PROXIMITY_MAX_MM = 500.0     # Distance for full speed
AUTOPILOT_PROXIMITY_MIN_MM = 100.0     # Distance for minimum speed
```

Vision Processing

```
YOLO_CONFIDENCE_THRESHOLD = 0.5        # Discard low-confidence detections
YOLO_INPUT_SIZE = 640                  # Model input resolution
YOLO_GPU_DEVICE = 0                    # CUDA device index
```

State Machine

```
SM_DELAY_SEARCH = 0.5                  # Delay before search handler (s)
SM_DELAY_GRASP = 1.0                   # Delay before grasp handler (s)
SM_MAX_GRASP_RETRIES = 1               # Grasp retry attempts
```

xArm Grasping

```
GRIPPER_OPEN_COUNT = 211.0             # Servo count for open
GRIPPER_CLOSE_COUNT = 687.0            # Servo count for closed
GRIPPER_BLOCK_TOLERANCE = 100.0        # Count gap indicating grasp
```

Modifying Parameters

1. **Edit the constants file**: `bash nano src/system_manager_package/system_manager_package/constants.py`
2. **Rebuild the workspace** (required for changes to take effect): `bash colcon build --symlink-install source install/setup.bash`
3. **Restart the affected nodes** or relaunch the full system.

Runtime Parameter Tuning

Some parameters can be tuned at runtime using the WSKR Dashboard:

- **Speed scale**: Adjust via slider (0.0 - 1.0)
- **Proximity limits**: Set min/max distances in mm
- **Model selection**: Pick different trained policies

Floor Detection Tuning

Use the Floor Tuner tool to interactively tune floor detection parameters:

```
ros2 run utilities floor_tuner
```

The tuner shows: - Live camera view - Floor mask preview - Sliders for blur, color threshold, gradient threshold - Save/Load buttons for parameter presets

Heading Tuner

Tune heading estimation parameters:

```
ros2 run utilities heading_tuner
```

Displays camera view with heading overlay and meridian visualization.

Launch File Arguments

Most launch files accept runtime arguments:

robot_bringup.launch.py

```
# Launch with GUI
ros2 launch system_manager_package robot_bringup.launch.py launch_gui:=true

# Set ArUco marker ID for box search
ros2 launch system_manager_package robot_bringup.launch.py search_aruco_id:=5

# Adjust vision sampling rate
ros2 launch system_manager_package robot_bringup.launch.py vision_sample_hz:=1.0
```

search_behavior.launch.py

```
# Adjust wander speed
ros2 launch wskr_search_behavior search_behavior.launch.py wander_speed_m_s:=0.15

# Change look duration
ros2 launch wskr_search_behavior search_behavior.launch.py look_duration_sec:=3.0

# Set confidence threshold
ros2 launch wskr_search_behavior search_behavior.launch.py confidence_threshold:=0.6
```

vision_processing.launch.py

```
# Change camera rate
ros2 launch vision_processing_package vision_processing.launch.py camera_rate_hz:=15.0

# Select GPU device
ros2 launch vision_processing_package vision_processing.launch.py gpu_device:=0

# Disable bbox-to-xyz service
ros2 launch vision_processing_package vision_processing.launch.py launch_bbox_to_xyz:=false
```

Troubleshooting

Common Issues and Solutions

Camera Not Publishing

Symptom: No images on `/camera1/image_raw/compressed`

Solutions: 1. Check camera device exists: `bash ls -la /dev/video*` 2. Verify camera is not in use by another process: `bash fuser /dev/video0` 3. Test camera with standard tools: `bash v4l2-ctl --device=/dev/video0 --list-formats-ext` 4. Check GStreamer pipeline: `bash gst-launch-1.0 v4l2src device=/dev/video0 ! jpegenc ! filesink location=test.jpg`

YOLO Not Detecting Objects

Symptom: Empty detections on `/vision/yolo/detections`

Solutions: 1. Verify TensorRT engine exists: `bash ls -la src/vision_processing_package/models/*.engine` 2. Check GPU utilization: `bash tegrastats # On Jetson` 3. Lower confidence threshold temporarily: `python # In constants.py YOLO_CONFIDENCE_THRESHOLD = 0.3` 4. Ensure good lighting and clear view of objects

Whisker Distances Always Zero

Symptom: `/WSKR/whisker_lengths` shows all zeros

Solutions: 1. Check floor detection is working: `bash ros2 topic hz /wskr_overlay/compressed` 2. Verify floor params are loaded: `bash ros2 param get /wskr_floor floor_blur_kernel` 3. Check camera is seeing the floor (not pointed at ceiling) 4. Adjust floor detection thresholds in Floor Tuner

Robot Not Moving

Symptom: cmd_vel published but robot stationary

Solutions: 1. Check Arduino connection: `bash ls -la /dev/ttyACM*` 2. Verify serial port permissions: `bash sudo usermod -a -G dialout $USER # Then logout and login again` 3. Check mecanum_serial_bridge output for errors 4. Test manual teleoperation: `bash ros2 run utilities mecanum_teleop`

State Machine Stuck

Symptom: Robot stays in one state and won't transition

Solutions: 1. Check action servers are available: `bash ros2 action list` 2. Verify required topics are publishing: `bash ros2 topic list ros2 topic hz /robot_state` 3. Check for error messages in state_manager_node output 4. Send "idle" command to reset: `bash ros2 topic pub /robot_command std_msgs/msg/String "{data: 'idle'}"`

xArm Not Grasping

Symptom: Grasp action completes but object not picked up

Solutions: 1. Check gripper servo positions: `python GRIPPER_OPEN_COUNT = 211.0 GRIPPER_CLOSE_COUNT = 687.0` 2. Verify object is within reach (check approach success) 3. Adjust gripper close tolerance: `python GRIPPER_BLOCK_TOLERANCE = 100.0 # Increase if false positives` 4. Test gripper manually: `bash ros2 service call /open_gripper_service std_srvs/srv/Trigger`

Foxglove Connection Issues

Symptom: Cannot connect to Foxglove WebSocket

Solutions: 1. Verify bridge is running: `bash ros2 node list | grep foxglove` 2. Check port is listening: `bash netstat -tlnp | grep 8765` 3. Ensure firewall allows the port: `bash sudo ufw allow 8765` 4. Try localhost connection first: `ws://localhost:8765`

High CPU Usage

Symptom: Jetson fans spinning loudly, system sluggish

Solutions: 1. Reduce camera publish rate: `bash ros2 launch vision_processing_package vision_processing.launch.py camera_rate_hz:=5.0` 2. Lower YOLO inference rate in constants.py 3. Use throttled topics in Foxglove: `python FOXGLOVE_CAMERA_THROTTLE_HZ = 2.0 FOXGLOVE_OVERLAY_THROTTLE_HZ = 5.0` 4. Monitor with tegrastats: `bash tegrastats`

ArUco Detection Not Working

Symptom: ArUco markers not detected in approach

Solutions: 1. Verify marker ID matches: `bash # In dashboard, enter correct ID` 2. Check marker is in camera field of view 3. Ensure good lighting (no glare on marker) 4. Verify ArUco dictionary matches: `python SEARCH_ARUCO_DICT = 'DICT_4X4_50'` 5. Test with larger marker or closer distance

Getting Help

1. **Check ROS 2 logs**: `bash ros2 log list ros2 log show <node_name>`
2. **Enable debug logging**: `bash ros2 run <package> <node> --ros-args --log-level debug`
3. **Record a rosbag for analysis**: `bash ros2 bag record -a`

4. **Visualize in Foxglove or RViz**: `bash ros2 run foxglove_bridge foxglove_bridge_launch.xml`

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